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United States Patent	12411720
Kind Code	B2
Date of Patent	September 09, 2025
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Cross-cluster load balancer

Abstract

Various examples are disclosed for performing cross-cluster load balancing. In some aspects, a workload is selected for cross-cluster migration. A destination cluster is identified for a migration of the workload from a source cluster to the destination cluster. A cross-cluster migration recommendation is generated to migrate the workload from the source cluster to the destination cluster.

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Appl. No.: 18/363656

Filed: August 01, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230376359 A1	Nov. 23, 2023

Related U.S. Application Data

continuation parent-doc US 16887932 20200529 US 11755385 child-doc US 18363656

Publication Classification

Int. Cl.: G06F9/50 (20060101); **G06F9/48** (20060101); **G06F9/54** (20060101)

U.S. Cl.:

CPC **G06F9/5088** (20130101); **G06F9/4875** (20130101); **G06F9/5077** (20130101);
G06F9/547 (20130101); G06F2209/505 (20130101)

Field of Classification Search

CPC: G06F (9/5088); G06F (9/4875); G06F (9/5077); G06F (9/547); G06F (2209/505); G06F (2209/501); G06F (9/4856)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application that claims priority to and the benefit of U.S. Non-provisional patent application Ser. No. 16/887,932, entitled “CROSS-CLUSTER LOAD BALANCES” and filed on May 29, 2020, which is incorporated by reference as if set forth herein in its entirety.

BACKGROUND

(1) Computer virtualization relates to the creation of a virtualized version of a physical device, such as a server, a storage device, a central processing unit (CPU), a graphics processing unit (GPU), or

other computing resources. Virtual machines, containers, and other workloads can perform enterprise tasks such as providing virtual desktop services, machine learning services, web services, and other services. The virtual machines can be executed using physical hardware in a datacenter. The datacenter can include many physical resources arranged in a cluster or group of hosts.

(2) Resource schedulers can determine the optimal placement of virtual machines and other workloads within a cluster. A resource scheduler can dynamically monitor a cluster and allocate physical resources to virtual machines within the cluster. The resource scheduler can also redistribute the virtual machines for load balancing purposes within the cluster. As datacenters expand, multi-cluster configurations are becoming more common. Single cluster private cloud configurations can be expanded to include public cloud resources to create a hybrid cloud environment.

(3) Resource management in a complex cloud environment with multiple clusters is a hard problem in view of scalability requirements, the heterogeneity of resource types, and the variability of workloads. Cluster-based limitations of existing solutions can result in a failure to effectively manage multi-cluster datacenter configurations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, with emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1 is a drawing of an example of a networked environment that includes a number of compute clusters that are managed using a management service that includes a cross-cluster load balancer.

(3) FIG. 2 is a drawing that illustrates an example of functionality implemented by the cross-cluster load balancer and other components of the networked environment of FIG. 1.

(4) FIG. 3 is a flowchart illustrating an example of functionality implemented by the cross-cluster load balancer and other components of the networked environment of FIG. 1.

(5) FIG. 4 is another flowchart illustrating an example of functionality implemented by the cross-cluster load balancer and other components of the networked environment of FIG. 1.

DETAILED DESCRIPTION

(6) The present disclosure relates to cross-cluster load balancing. Resource schedulers can dynamically monitor datacenter clusters, allocating and redistributing workloads to the physical resources. As datacenters expand, multi-cluster configurations are becoming more common. Private cloud configurations can be expanded to include public cloud resources, creating a hybrid cloud environment. Existing technologies can fail to effectively manage multi-cluster datacenter configurations. However, the present disclosure describes mechanisms that can more efficiently and effectively manage workloads across multi-cluster datacenters.

(7) With reference to FIG. 1, an example of a networked environment **100** is shown. The networked environment **100** can include a management environment **103** and various clusters **106** in communication with one another over a network **109**. The network **109** can include, for example, the Internet, intranets, extranets, wide area networks (WANs), local area networks (LANs), wired networks, wireless networks, other suitable networks, or any combination of two or more such networks. For example, the networks can include satellite networks, cable networks, Ethernet networks, telephony networks, and other types of networks. As the networked environment **100** can serve up virtual desktops to end users, the networked environment **100** can also be described as a

virtual desktop infrastructure (VDI) environment.

(8) In various embodiments, each of the clusters **106** can include a plurality of physical devices including the hosts **115a . . . 115m** (the hosts **115**), installed in racks **112a** and **112b** (the racks **112**) which can make up a server bank, aggregate computing system, or a computer bank of a datacenter. The cluster **106** can include a set of hosts **115** and other hardware resources that are managed as a group by a cluster scheduling service **137** that operates at a cluster level to balance workloads **145** within a single cluster **106**. The various physical devices or hosts **115** of a cluster **106** can be located in a single facility or can be located in a separate geographical location. Multiple clusters **106** can be located in a single facility, and additional clusters **106** can be located in a separate geographical location. The devices in the clusters **106** can include any number of physical machines, virtual machines, virtual appliances, and software, such as operating systems, drivers, hypervisors, scripts, and applications. The various clusters **106** can include private cloud, public cloud, and hybrid cloud configurations.

(9) In various examples, when a physical computing device or host **115** is added to a cluster **106**, the host **115** can be configured to access a datastore **130** corresponding to that cluster **106**. In other words, in order to access a datastore **130**, a host **115** must be a member of the corresponding cluster **106**. In some cases, the cluster **106** can include a virtual storage area network (vSAN) cluster in a VMware vSphere® or similar environment. A vSAN cluster **106** can aggregate capacity devices of the vSAN cluster **106** to create a single storage pool shared across all hosts **115** in the vSAN cluster **106**. For example, the datastore **130** can include a vSAN datastore **130** that aggregates the storage available to all of the physical resources and hosts **115** that are assigned to the cluster **106**.

(10) The datastore **130** can include storage area network, network file system, virtual machine file system, and hyperconverged infrastructure storage. In some examples the datastore **130** can be cluster-specific so that the datastore **130** is dedicated to a single cluster **106** that has direct access to that cluster-level datastore **130**. The datastore **130** can be accessible to all hosts **115** of a particular cluster **106**. Alternatively, the datastore **130** can be accessible to the hosts **115** of multiple clusters **106**.

(11) A hypervisor can be installed on a host **115** to support workload execution space within which one or more virtual machines can be concurrently instantiated and executed. In some examples, the hypervisor can include the VMware ESX™ hypervisor, the VMware ESXi™ hypervisor, or similar hypervisor. The cluster scheduling services **137** can manage virtual machines, containers, and other workloads **145a . . . 145f** (workloads **145**) to the hosts **115** at a cluster level. A respective cluster scheduling service **137** can manage a particular cluster **106**. The cluster scheduling service **137** can monitor, assign, and redistribute workloads **145** among the hosts **115** in its cluster **106**. For example, a cluster scheduling service **137** can manage workloads **145** among the hosts **115** that are assigned to a particular cluster **106** or a datastore **130**.

(12) Additionally, if a host **115** includes an instance of a virtual machine, container, or another workload, the workload can be referred to as a “guest.” It is understood that the clusters **106** are scalable, meaning that the clusters **106** in the networked environment **100** can be dynamically scaled up and scaled down to include additional and fewer hosts **115**, switches **118**, accelerators, power sources, and other components, without degrading performance of the virtualization environment. For example, the cluster scheduling service **137** can include an automatic scaling functionality that dynamically scales up and scales down a particular cluster **106**.

(13) The management service **135** can also include a cross-cluster load balancer **139** capable of cross-cluster migration of workloads **145** from one cluster **106** to another. The cross-cluster load balancer **139** can operate at a datacenter level for all clusters **106** and can perform cross-cluster migrations for load balancing, cost saving, and other purposes. The migration can include no downtime or minimal downtime of the workloads **145** during migration. For example, the workload **145** can be copied from a source host **115**, and replicated to a destination host **115**. Once the workload **145** is executing on the destination host **115**, the workload **145** can be halted and

removed from the source host **115**. Unlike traditional technologies that might extend a cluster **106** or a datastore **130** across multiple sites, and rely on a cluster scheduling service **137** for load balancing, the cross-cluster load balancer **139** is capable of load balancing by workload migration across multiple clusters **106**.

(14) The cross-cluster load balancer **139** can provide a number of internal application programming interfaces (APIs), including a recommendation API, a migration API, and a stop or cancel API. The recommendation API can be invoked with input parameters including a set of participant clusters **106** to analyze for cross-cluster load balancing, and a mode selection between manual or automated modes. The participant set of clusters **106** can include all clusters **106** of an enterprise datacenter, or a manually-selected subset of the clusters **106**.

(15) In manual mode, the cross-cluster load balancer **139** can generate and provide migration recommendations for manual acceptance. The migration API can be invoked if the management service **135** identifies a manual indication to implement the migration recommendations. The cancel API can be invoked to manually stop a cross-cluster migration that is being performed. The cancel API can also clean up or delete partially-deployed workloads **145** from the prospective destination. In automated mode, the cross-cluster load balancer **139** can generate migration recommendations and perform migrations based on the recommendations.

(16) A decision made by the cluster scheduling service **137** to scale up or scale down a cluster **106** can conflict with the cross-cluster migration decisions of the cross-cluster load balancer **139**. Likewise, a decision made by the cross-cluster load balancer **139** to scale workloads **145** of a cluster **106** by cross-cluster migration can conflict with the automatic host scaling functionality of the cluster scheduling service **137**. For example, if cluster resources are overloaded, the cluster scheduling service **137** can decide to add a host **115** to a cluster **106**, while the cross-cluster load balancer **139** can decide to move out several workloads **145** from the cluster. To avoid a potential conflict, the management service **135** provides a mutual exclusion control rule over a cluster **106** between the cross-cluster load balancer **139** and the cluster scheduling service **137**. While both the cross-cluster load balancer **139** and the cluster scheduling service **137** can be concurrently enabled for a single cluster **106**, the mutual exclusion control rule can indicate that when the cross-cluster load balancer **139** (or the cluster scheduling service **137**) works on a cluster **106**, the cluster scheduling service **137** (or the cross-cluster load balancer **139**) is temporarily prevented from making scaling decisions.

(17) In one example, if the cross-cluster load balancer **139** migrates a workload **145** from one cluster **106** to another cluster **106**, this can increase resource usage of the destination cluster. However, the mutual exclusion control rule can prevent the cluster scheduling service **137** of the destination cluster **106** from scaling up or adding a host **115** to the destination cluster for a period of time. The cluster scheduling service **137** of the source cluster **106** can also be prevented from scaling down or removing a host **115** for a period of time.

(18) The cross-cluster load balancer **139** can provide an indication that it is generating recommendations for cross-cluster migrations or has generated recommendations for cross-cluster migrations, and the cluster scheduling service **137** can be temporarily prevented from making scaling decisions. The cluster scheduling service **137** can be prevented from making scaling decisions in response to various events associated with the cross-cluster load balancer **139** working on a cluster **106**, including generating recommendations, providing a cross-cluster migration recommendation, or completion of a cross-cluster migration according to the mutual exclusion control rule. The cross-cluster load balancer **139** can be prevented from generating, providing, and performing cross-cluster migrations in response to various events associated with the cluster scheduling service **137** working on a cluster **106**, including generating, providing, and performing host scaling recommendations.

(19) The hosts **115** in the cluster **106** are monitored by the management service **135** and, in the event of a failure, the virtual machines, virtual appliances, or workloads **145** on a failed host **115**

are restarted on alternate hosts **115**. In some examples, the cluster **106** can include a high-availability cluster or other computing cluster. A high-availability cluster can include a group of computing devices that act as a single system and provide a continuous uptime. Some of the hosts in the cluster **106** can be designated as primary hosts **115**, and other hosts in the cluster **106** can be designated as secondary hosts. The primary hosts **115**, for example, can maintain and replicate states of the cluster **106** and can be used to initiate failover actions. Any host **115** that joins the cluster **106** can communicate with a host **115**, such as an existing primary host **115**, to complete its configuration.

(20) The devices in the racks **112** can include, for example, memory and storage devices, hosts **115a . . . 115m**, switches **118a . . . 118d**, graphics cards with GPUs, artificial intelligence and other accelerators, central processing units (CPUs), power supplies, and similar devices. The devices, such as hosts **115** and switches **118**, can have dimensions suitable for quick installation in slots **124a . . . 124i** on the racks **112**. In various examples, the hosts **115** can include requisite physical hardware and software to create and manage a virtualization infrastructure. The physical hardware resources for a host **115** can include a CPU, data bus, memory, graphics card/accelerator, artificial intelligence accelerator, other accelerators, and other components. The host memories can include short-term access memories including random access memory (RAM), dynamic RAM (DRAM), and synchronous DRAM (SDRAM); and long term data storage including solid state drives (SSDs), and hard disk drives (HDDs). In some examples, the hosts **115** can include a pre-configured hyper-converged computing device where a hyper-converged computing device includes pre-tested, pre-configured, and pre-integrated storage, server and network components, including software, that are positioned in an enclosure installed in a slot **124** on a rack **112**.

(21) The management environment **103** can include, for example, a server or any other system providing computing capability. Alternatively, the management environment **103** can include one or more computing devices that are arranged, for example, in one or more server banks, computer banks, clusters, or other arrangements. The management environment **103** can include a grid computing resource or any other distributed computing arrangement. The computing devices can be located in a single installation or can be distributed among many different geographical locations. Although shown separately from the clusters **106**, it is understood that in some examples the management environment **103** can include the clusters **106**.

(22) The management environment **103** can include or be operated as one or more virtualized computer instances. For purposes of convenience, the management environment **103** is referred to herein in the singular. Even though the management environment **103** is referred to in the singular, it is understood that a plurality of management environments **103** can be employed in the various arrangements as described above. As the management environment **103** communicates with the clusters **106** and client devices **108** for end users over the network **109**, sometimes remotely, the management environment **103** can be described as a remote management environment **103**, in some examples. Additionally, in some examples, the management environment **103** can be implemented in hosts **115** of a rack **112** and can manage operations of a virtualized computing environment. Hence, in some examples, the management environment **103** can be referred to as a management cluster in the clusters **106**.

(23) The datastore **130** can include a database or other memory that includes, for example, virtual appliance images, virtual machine images, other workload images and other data associated with the various workloads **145**. A workload image can include virtual machine images utilized by the components of the management service **135** in provisioning, executing, and migrating a workload **145** ready to run on a hypervisor of a host **115** in the cluster **106**.

(24) The components executed in the management environment **103** can include, for example, a cluster management service **135** as well as other applications, services, processes, systems, engines, or functionality not discussed in detail herein. The cluster management service **135** can be executed to oversee the operation of the networked environment **100** through management of the clusters

106, as well as the devices and software that make up the clusters **106**, such as virtual machines and virtual appliances. In some examples, an enterprise, organization, or other entity, can operate the cluster management service **135** to oversee or manage the operation of devices in the racks **112**, such as hosts **115**, switches **118**, accelerators, power supplies, cooling systems, and other components.

(25) As discussed above, the cluster management service **135** can include a number of cluster scheduling services **137** that operate at a cluster level, as well as a cross-cluster load balancer **139** that operates across all clusters **106** of the datacenter. The various physical and virtual components of the clusters **106** can process workloads **145a . . . 145f**. Workloads **145** can refer to the amount of processing that a host **115**, switch **118**, accelerator, or other physical or virtual component has been instructed to process or route at a given time. The workloads **145** can be associated with virtual machines or other software executing on the hosts **115**. The cluster management service **135** can maintain a listing of active or inactive workloads **145** as well as oversee the assignment of various workloads **145** to various devices in the clusters **106**. The workloads **145** can be routed to various hosts **115** by the switches **118** as network traffic **148a . . . 148b**.

(26) Referring next to FIG. **2**, shown is an example of the functionality implemented by the cross-cluster load balancer **139** and other components of the networked environment **100** of FIG. **1**. In this example, the networked environment **100** can include the cross-cluster load balancer **139** operating at a datacenter level to control cluster A and cluster B. Clusters A and B are used for clarity purposes to refer to various clusters **106**.

(27) Cluster A can include hosts A1 and A2, as well as a datastore A. Cluster B can include hosts B1 and B2, as well as a datastore B. The hosts A1, A2, B1, and B2 are used for clarity purposes to refer to various hosts **115**. The cluster-level datastores A and B are used for clarity purposes to refer to various cluster-level datastores **130**. The datastore A can include workloads A1, A2, and A3. The datastore B can include workloads B1 and B2. The workloads A1, A2, A3, B1, and B2 are used for clarity purposes to refer to various workloads **145**.

(28) In cluster A, host A1 can execute workload A1; host A2 can execute workload A2 as well as workload A3. The cluster scheduling service **137a** can manage intra-cluster load balancing and migrations for the workloads A1, A2, A3 among the hosts A1 and A2, as well as host scaling to include additional or fewer hosts within cluster A. Cluster statistics updater **245a** can collect and monitor cluster resource data **247a** for cluster A. In some cases, the cluster statistics updater **245a** can identify cluster resource data **247a** that is collected by the cluster scheduling service **137a**, and provide this cluster resource data **247a** to the cross-cluster load balancer **139**. The cluster resource data **247a** can include compute, memory, storage, network, accelerate, and other hardware resource information for cluster A.

(29) In cluster B, host B1 can execute workload B1; and host B2 can execute workload B2. The cluster scheduling service **137b** can manage intra-cluster load balancing and migrations for the workloads B1 and B2 among the hosts B1, B2, and B3, as well as host scaling to include additional or fewer hosts within cluster B. Cluster statistics updater **245b** can collect and monitor cluster resource data **247b** for cluster B. In some cases, the cluster statistics updater **245b** can monitor cluster resource data **247b** that is collected by the cluster scheduling service **137b**, and can provide this cluster resource data **247b** to the cross-cluster load balancer **139**. The cluster resource data **247b** can include compute, memory, storage, network, accelerate, and other hardware resource information for cluster B.

(30) Compute resource information can include a measure of compute capacity provided, used, and percentage utilized. Compute capacity can be measured as a number of processors of a particular type, or a frequency of operations such as megahertz, gigahertz, or terahertz. Memory resource information can include a measure of temporary or access memory capacity provided, used, and percentage utilized. Memory capacity can be measured as a number of megabytes, gigabytes, or terabytes. Storage resource information can include a measure of data storage capacity provided,

used, and percentage utilized. Data storage capacity can be measured as a number of megabytes, gigabytes, or terabytes. Network resource information can include a measure of network communication capacity provided, used, and percentage utilized. Network communication capacity can be measured as a number of inputs/outputs (IOs), or a network bandwidth. Accelerate resource information can include a measure of accelerator-type-specific compute capacity provided, used, and percentage utilized for each hardware accelerator type. Accelerator types can include graphics processing accelerators, artificial intelligence accelerators, and other hardware accelerators. Accelerate capacity can be measured as a number of accelerators or accelerator processors of a particular type, or a frequency of operations such as megahertz, gigahertz, or terahertz. Percentage utilized for each hardware resource can refer to a current instantaneous reading or a calculated time-average.

(31) The cross-cluster load balancer **139** can monitor the cluster resource data **247a** and **247b** to determine whether to perform one or more cross-cluster migrations for load balancing. The cross-cluster load balancer **139** can generate cluster resource usages **261** for each of the clusters A and B, as well as other clusters **106**. The cluster resource usage **261** can include a cluster resource usage score that considers one or more of the compute, memory, storage, network, accelerate, and other hardware resource information for each cluster. The cluster resource usage **261** can refer to cluster-level resource usage.

(32) The cluster resource usage **261** can be based on percent resource usage or resource contention. Percent resource usage can refer to the percentage of the total resources that are utilized. Resource contention can refer to the amount of expected performance loss due to the contention for a resource as a percentage. For example, if the memory resource contention for the cluster is 80%, memory contention can cause 80% percent loss in overall cluster throughput across all workloads. Resource contention can be related to efficiency so that a resource contention of 80% corresponds to an efficiency of 20%. The cluster resource usage **261** can also be determined using a weighted average or other weighted cluster resource usage score can be calculated, with each resource being weighted with a predetermined coefficient.

(33) In some cases, the cluster resource usage **261** can refer to a dominant resource usage for each cluster. A dominant resource usage can refer to the hardware resource with the highest percent resource usage or resource contention. For example, if cluster A uses 50% of its storage capacity, 75% of its network capacity, and 90% of its compute capacity, its dominant resource usage can be based on compute capacity, and the cluster usage score can be based on the 90% compute capacity usage. If cluster B uses A uses 60% of its storage capacity, 35% of its network capacity, and 50% of its compute capacity, its dominant resource usage can be based on storage capacity, and the cluster usage score can be based on the 60% storage capacity usage. When sorting, the cross-cluster load balancer **139** can identify that 90% dominant resource usage for cluster A is greater than the 60% dominant resource usage for cluster B.

(34) The cross-cluster load balancer **139** can sort the clusters in a ranked list in descending order according to their cluster resource usage **261**. The cross-cluster load balancer **139** can select the highest cluster from the ranked list as a source cluster to analyze. In this case, cluster A can be the source cluster. The cross-cluster load balancer **139** can analyze each of the workloads of cluster A to generate cross-cluster migration recommendations **263**. In this example, the cross-cluster load balancer **139** can recommend a cross-cluster migration of the workload A3 to cluster B. The cross-cluster load balancer **139** can select the cluster B as a destination cluster based on the cluster resource usage **261** for cluster B. For example, the destination cluster B can have a lower cluster resource usage **261** than the source cluster A. In some cases, the destination cluster B can be selected based on having a lowest cluster resource usage **261** among the clusters **106** of a datacenter.

(35) The cross-cluster migration recommendation **263** can specify cluster B corresponding to datastore B, and can further specify the optimal host as host B3. In other examples, the cross-

cluster migration recommendation **263** can specify a migration to cluster B, and the cluster scheduling service **137b** can select a host **115** for the workload A3. The cross-cluster load balancer **139** or the management service **135** can generate a user interface that presents the cross-cluster migration recommendations **263** for manual approval. The cross-cluster load balancer **139** can also use the cross-cluster migration recommendations **263** to automatically migrate workloads **145**.

(36) The cross-cluster load balancer **139** can generate a workload source score **265** for the workload A3 based on its source cluster A, source datastore A, and host A2. In one example, the workload source score **265**, scores, can be calculated based on a hardware resource efficiency of the workload A3 executed on cluster A. The workload source score **265** can be based on the efficiency of host-specific resource usage and/or cluster-level resource usage for the source of workload A3. In other words, the workload source score **265** can refer to a source hardware resource efficiency associated with the workload A3. The workload source score **265** can be calculated based on equation 1, where $\text{eff}(\text{cpu.sub.s})$ can refer to source CPU efficiency, $\text{eff}(\text{mem.sub.s})$ can refer to source short term or access memory efficiency, $\text{eff}(\text{net.sub.s})$ can refer to source network efficiency, and $\text{eff}(\text{storage.sub.s})$ can refer to source storage efficiency.

$$\text{score.sub.s} = \text{eff}(\text{cpu.sub.s}) \times \text{eff}(\text{mem.sub.s}) \times \text{eff}(\text{net.sub.s}) \times \text{eff}(\text{storage.sub.s}) \quad (1)$$

(37) Efficiency can represent the amount of expected performance loss due to the contention for a resource, assuming all other resources are in perfect state, and the unit of the efficiency is percent. For example, if the memory efficiency is 80%, memory contention can cause 20% loss in overall workload or virtual machine throughput. In the example of equation 1, the overall workload source score **265**, score.sub.s , can be a product of the efficiencies of the resources considered. Other measures, such as weighted products or weighted averages can also be used, where each resource is weighted according to a predetermined coefficient based on subjective or objective performance criteria.

(38) The cross-cluster load balancer **139** can generate workload destination scores **267** or candidate destination scores for the workload A3 based on candidate destination cluster B. A workload destination score **267** can be based on efficiency of cluster-level resource usage for the candidate destination cluster B for workload A3, or the efficiency of host-specific resource usage if the workload A3 was assigned to a particular candidate host among the hosts B1, B2, and B3. The workload destination score **267** can be based on the efficiency of host-specific resource usage and/or cluster-level resource usage for the candidate destination for workload A3. In other words, the workload destination score **267** can refer to a destination hardware resource efficiency associated with the workload A3. The workload destination score **267**, score.sub.d , can be calculated based on equation 2.

$$\text{score.sub.d} = \text{eff}(\text{cpu.sub.d}) \times \text{eff}(\text{mem.sub.d}) \times \text{eff}(\text{net.sub.d}) \times \text{eff}(\text{storage.sub.d}) \quad (2)$$

In equation 2, $\text{eff}(\text{cpu.sub.d})$ can refer to candidate destination CPU efficiency, $\text{eff}(\text{mem.sub.d})$ can refer to destination short term or access memory efficiency, $\text{eff}(\text{net.sub.d})$ can refer to destination network efficiency, and $\text{eff}(\text{storage.sub.d})$ can refer to destination storage efficiency.

(39) The cross-cluster load balancer **139** can further determine whether to make a cross-cluster migration recommendation **263** based on cross-cluster migration gains **269**. The cross-cluster migration gain **269** can refer to the hardware resource efficiency benefit of moving the workload A3 from its source cluster, datastore, and host to a destination cluster, datastore, and host. The cross-cluster migration gain **269** can be determined based on a gain rate and a gain duration for a cross-cluster migration. The gain rate and gain duration can be determined based on host-specific resource measures or cluster-level resource measures. For example, the gain rate can be calculated according to equation 3 based on CPU demand or estimated requirement for the workload A3.

$$\text{gainrate} = \text{cpuDemand} \times (\text{score.sub.d} - \text{score.sub.s}) \quad (3)$$

While equation 3 uses the CPU demand, another hardware requirement can also be utilized alternatively or additionally to the CPU demand.

(40) The gain duration can refer to how long the gain rate is predicted or estimated to last on the

destination side. We estimate the gain duration based on the workload in/out migration behavior on each host **115** and the cross-cluster migration time. The in/out migration behavior can be based on cross-cluster migrations, but can also consider intra-cluster migrations. If a host **115** has more frequent in/out actions, the gain duration value is reduced. The estimated cross-cluster migration time is also taken out of the gain duration since there is no performance gain during migration.

(41) The cross-cluster load balancer **139** can determine the cross-cluster migration gain **269** as the product of a gain rate and a gain duration. The cross-cluster load balancer **139** can compare the cross-cluster migration gain **269** to an estimated or predetermined cross-cluster migration cost **271**, as shown in equation 4.

$$\text{gainrate} \times \text{gainDuration} > \text{ClusterMigrationCost} \quad (4)$$

The cross-cluster migration cost **271** can be determined based on measured values in live or simulated environments. If the product of the gain rate and the gain duration is greater than the cross-cluster migration cost **271**, then the cross-cluster load balancer **139** can make or provide a cross-cluster migration recommendation **263** that specifies a cross-cluster migration of the workload **A3** to the cluster B and/or the host **B3**. In an automated mode, the cross-cluster load balancer **139** can automatically perform the migration. In a manual mode, the cross-cluster load balancer **139** can present the cluster migration recommendation **163** for manual acceptance or rejection through a console user interface of the management service **135**.

(42) The cross-cluster load balancer **139** can consider workload-to-workload affinity rules, including affinities and anti-affinities. And affinity rule can specify that workloads should be (or should not be) executed on the same host or same cluster. If the workload **A3** has an affinity with the workload **A2**, the cross-cluster load balancer **139** can consider the two workloads and migrate the workloads **A2** and **A3** to cluster B if both workloads **A2** and **A3** have cross-cluster migration gains **269** that are greater than their cross-cluster migration costs **271**.

(43) While cross-cluster migration recommendations **263** can be generated and implemented based on hardware efficiencies, the cross-cluster load balancer **139** can also perform cross-cluster live migrations in response to a number of other scenarios. For example, if the workload **A3** has an affinity with workload **B1**, the cross-cluster load balancer **139** can identify this, and can migrate the workload **A3** from cluster A to cluster B to enforce the affinity rule. This can be performed regardless of hardware resource efficiency. In further examples, workload **B1** can maintain information in a database or data structure that is utilized by a number of other workloads including the workload **B2** and the workload **A3**. The cross-cluster load balancer **139** can identify that the workload **A3** includes an application that uses the database maintained by workload **B1**, and can migrate workload **A3** to cluster B to improve application performance. The cross-cluster load balancer **139** can for example, the cross-cluster load balancer **139** can detect that the host **A2**, or the cluster A has been manually placed in a maintenance mode. The cross-cluster load balancer **139** can recommend and implement cross-cluster migrations of the workloads **A1**, **A2**, and **A3** to maintain accessibility.

(44) Turning now to FIG. 3, shown is a flowchart **300** illustrating an example of functionality implemented by a management service **135** with a cross-cluster load balancer **139**, and other components of the networked environment **100** of FIG. 1. The steps of the flowchart **300** can depict an example of various executional states implemented by the cross-cluster load balancer **139**, according to one or more examples. The separation or segmentation of functionality as discussed herein is presented for illustrative purposes only.

(45) In step **303**, the cross-cluster load balancer **139** can be in a ready state. In the ready state, the cross-cluster load balancer **139** can be initialized and ready for invocation. The management service **135** can initialize the cross-cluster load balancer **139** within the multi-cluster management environment **103**.

(46) In step **306**, the cross-cluster load balancer **139** can generate cross-cluster migration recommendations **263** as discussed with respect to FIG. 2. This process is also discussed in further

detail with respect to FIG. 4. The cross-cluster load balancer **139** can receive an API call that invokes the recommendation API. The cross-cluster load balancer **139** can analyze the clusters **106** to migrate workloads **145** from clusters **106** with higher resource utilization, and to clusters **106** with lower resource utilization.

(47) The cross-cluster load balancer **139** can analyze the workloads **145** from a high-utilization cluster **106** in order to identify whether or not each of the workloads **145** can benefit from a cross-cluster migration. The cross-cluster load balancer **139** can determine a cross-cluster migration gain **269** for each workload **145** in the high-utilization cluster **106**. The cross-cluster migration gain **269** can be determined using host-specific and/or cluster-level resource usage. If the cross-cluster migration gain **269** is greater than a cross-cluster migration cost **27**, the cross-cluster load balancer **139** can accept the cross-cluster migration recommendation **263**.

(48) The cross-cluster migration recommendation **263** can identify a destination cluster **106**, and can further specify a destination datastore **130**, and a specific host **115** that are associated with a maximal cross-cluster migration gain **269**. If an exception or error occurs, the process can move to step **303**. Otherwise, if cross-cluster migration recommendations **263** are generated, then the process can move to step **309**.

(49) In step **309**, the cross-cluster load balancer **139** can propose the cross-cluster migration recommendations **263** for authorization. The cross-cluster load balancer **139** can reserve resources in the destination cluster **106**, for example, within the destination datastore **130** and host **115**. In manual mode, the cross-cluster load balancer **139** can transmit or otherwise provide the cross-cluster migration recommendations **263** to a user for manual acceptance or rejection. The management service **135** can include a console user interface that presents the cross-cluster migration recommendations **263**, identifying the workload **145**, the source and destination clusters **106** and hosts **115**, as well as the expected gain.

(50) The cross-cluster migration recommendations **263** can be associated with a timeout period. If the cross-cluster migration recommendations **263** are unauthorized within the timeout period, the process can move to step **312**. Otherwise, if the cross-cluster migration recommendations **263** are authorized, then the process can move to step **315**. A user can alternatively request that the cross-cluster load balancer **139** reevaluate the clusters and generate new cross-cluster migration recommendations **263**, and the process can move to step **306**. In automated mode, the cross-cluster load balancer **139** can automatically authorize the cross-cluster migration recommendations **263** and move to step **315**.

(51) In step **315**, the cross-cluster load balancer **139** can perform cross-cluster migrations. For example, the cross-cluster load balancer **139** can work with the cluster scheduling service **137** to prepare the workloads **145** for execution on the destination cluster **106**, datastore **130**, and host **115**. The cross-cluster load balancer **139** can perform a number of cross-cluster migrations according to the authorized cross-cluster migration recommendations **263**. If a cross-cluster migration fails or succeeds, the process can move to step **303**. The cross-cluster migration can fail if it takes longer than a threshold migration time, if the destination host **115** or datastore **130** goes offline, or other exceptions occur. In the case of a failure, the cross-cluster load balancer **139** can clean up or delete partially-deployed workloads **145** from the destination cluster **106**, and free the resources that were reserved for migration. In some cases, the cross-cluster load balancer **139** can provide a progress indicator in the console user interface of the management service **135**. The console user interface can also provide a user interface element that, when selected, cancels the migration. If the migration is manually cancelled, the process can move to step **318**.

(52) In step **318**, the cross-cluster load balancer **139** can cancel the cross cluster migration. The cancel API can be invoked to manually stop a cross-cluster migration that is being performed. The cancel API can also clean up or delete partially-deployed workloads **145** from the destination cluster **106**, and free the resources that were reserved for migration. Thereafter, the process can move to step **303**.

(53) FIG. 4 shows a flowchart **400** that provides one example of functionality implemented by a management service **135** with a cross-cluster load balancer **139**, and other components of the networked environment **100** of FIG. 1. The flowchart **400** can depict an example of how the cross-cluster load balancer **139** can analyze the clusters **106** to generate cross-cluster migration recommendations **263**, according to one or more examples. The separation or segmentation of functionality as discussed herein is presented for illustrative purposes only.

(54) In step **403**, the cross-cluster load balancer **139** can sort clusters **106** based on cluster resource usage **261**. For example, the cross-cluster load balancer **139** can receive a request to generate cross-cluster migration recommendations **263**. Such a request can include invoking the recommendation API with input parameters including a set of participant clusters **106** to analyze for cross-cluster load balancing. The participant set of clusters **106** can include all clusters **106** of an enterprise datacenter, or a manually-selected subset of the clusters **106**. The request can also indicate a mode selection of manual mode or automated mode.

(55) The cross-cluster load balancer **139** can analyze the set of participant clusters **106** to determine a cluster resource usage **261** for each cluster **106**. The cluster resource usage **261** can include a cluster resource usage score that considers one or more of the compute, memory, storage, network, accelerate, and other hardware resource information for each cluster. The cluster resource usage **261** can refer to a dominant resource usage for each cluster **106**, an average resource contention for one or more of the hardware resources, or another measure. The cross-cluster load balancer **139** can generate a ranked list of the participant clusters **106** in descending order of their cluster resource usage **261**.

(56) In step **406**, the cross-cluster load balancer **139** can identify or select a source cluster **106**. The source cluster **106** can be the cluster **106** that has the highest cluster resource usage **261** of the participant clusters **106**. For example, the source cluster **106** can be ranked highest in the ranked list of the participant clusters **106**.

(57) In step **409**, the cross-cluster load balancer **139** can analyze the workloads **145** of the source cluster **106**. Generally, the cross-cluster load balancer **139** can go through each of the workloads **145** of the source cluster **106** to determine whether to perform a cross-cluster migration of that workload **145** from the high-usage source cluster **106** to a lower-usage destination cluster **106**.

(58) In step **412**, the cross-cluster load balancer **139** can select a workload **145** to analyze for cross-cluster migration. The cross-cluster load balancer **139** can select the workload **145** at random or based on its hardware resource requirements, execution time, and other factors.

(59) In step **415**, the cross-cluster load balancer **139** can select a destination cluster **106**. The destination cluster **106** can be a cluster from a subset of the participant clusters **106** with the source cluster **106** removed. In some cases, the cross-cluster load balancer **139** can select the destination cluster **106** based on its cluster resource usage **261**. For example, the destination cluster **106** can be the lowest-ranked cluster in the ranked list of the participant clusters **106**. The cross-cluster load balancer **139** can select a destination cluster **106** that has a lower cluster resource usage **261** than the source cluster **106**. Since a datastore **130** aggregates the storage resources of a cluster **106**, the destination cluster selection can refer to a selection of the destination datastore **130**.

(60) In step **418**, the cross-cluster load balancer **139** can select a destination host **115**. In some cases, the cross-cluster load balancer **139** can select the destination cluster **106** based on its cluster resource usage **261**. Otherwise, the cross-cluster load balancer **139** can try various hosts **115** in succession to determine an optimal destination host **115** for the migration. In some cases, the cross-cluster load balancer **139** can communicate the resource requirements of the workload **145** to the cluster scheduling service **137**, and the cluster scheduling service **137** can identify the destination host **115**.

(61) In step **421**, the cross-cluster load balancer **139** can determine whether a cross-cluster migration gain **269** is greater than a cross-cluster migration cost **271**. The cross-cluster migration gain **269** can refer to the resource benefit of moving a workload **145** from its source cluster **106** to a

destination cluster **106**. The cross-cluster migration gain **269** can be determined based on host-specific or cluster-level measures, among the various embodiments. The cross-cluster load balancer **139** can determine the cross-cluster migration gain **269** as the mathematical product of a gain rate and a gain duration. The cross-cluster load balancer **139** can compare the cross-cluster migration gain **269**, to a cross-cluster migration cost **271**, as described in equation 4. If the cross-cluster migration gain **269** is greater than the cross-cluster migration cost **271**, then the cross-cluster load balancer **139** can generate a cross-cluster migration recommendation **263** that specifies the destination cluster **106**, as well as the destination datastore **130** and destination host **115**.

(62) The cross-cluster load balancer **139** can perform an iterative process to determine the best destination cluster **106**, including the destination datastore **130** and destination host **115**. For example, if the cross-cluster migration gain **269** is greater than the cross-cluster migration cost **271**, the cross-cluster load balancer **139** can compare the current cross-cluster migration gain **269** to a previous best cross-cluster migration gain **269** based on another destination datastore **130** and destination host **115**.

(63) In step **424**, the cross-cluster load balancer **139** can determine if a maximum number of cross-cluster migration recommendations **263** is met. The number of migrations can be limited to a predetermined number of recommendations. If the total number of cross-cluster migration recommendations **263** is equal to the maximum number, then the process can move to step **427**. Otherwise, if the total number of cross-cluster migration recommendations **263** is less than the predetermined maximum, then the process can move step **430**.

(64) In step **427**, the cross-cluster load balancer **139** can provide the cross-cluster migration recommendations **263** or automatically migrate workloads **145** according to the cross-cluster migration recommendations **263**. In automated mode, the cross-cluster load balancer **139** can automatically migrate workloads **145** according to the cross-cluster migration recommendations **263**. In manual mode, the cross-cluster load balancer **139** can transmit or otherwise provide the cross-cluster migration recommendations **263** to the management service **135**. The management service **135** can generate a user interface that includes the cross-cluster migration recommendations **263**. The user interface can also include a description or identification of the workload **145**, as well as resource usage information for the source cluster **106** and the destination cluster **106**. The resource usage information can include current cluster resource usage **261** and projected cluster resource usage if the migration is carried out.

(65) In step **430**, the cross-cluster load balancer **139** can determine whether there are additional hosts **115** that have not been analyzed as destination candidates for the workload **145**. If there are additional hosts **115**, then the cross-cluster load balancer **139** can move to step **418** and select another candidate destination host. The cross-cluster load balancer **139** can iteratively determine cross-cluster migration gains **269** for each host **115** of the current candidate destination cluster **106**, and can identify the host **115** that is associated with the highest cross-cluster migration gain **269**. If there are no additional hosts **115**, then the cross-cluster load balancer **139** can move to step **433**.

(66) In step **433**, the cross-cluster load balancer **139** can determine whether there are additional destination clusters **106** that have not been analyzed as destination candidates for the workload **145**. If there are additional destination clusters **106**, then the cross-cluster load balancer **139** can move to step **415** and select another candidate destination cluster **106**. As a result, the cross-cluster load balancer **139** can iteratively select each destination cluster **106** to determine the optimal destination for the workload **145**. If there are no additional destination clusters **106**, then the cross-cluster load balancer **139** can move to step **436**.

(67) In step **436**, the cross-cluster load balancer **139** can determine whether the source cluster **106** includes additional workloads **145** that have not been analyzed for cross-cluster migration. If there are additional workloads **145**, then the cross-cluster load balancer **139** can move to step **412** and select another workload **145** from the source cluster **106**. As a result, the cross-cluster load balancer **139** can iteratively determine whether each workload **145** can benefit from cross-cluster migration.

If there are no additional workloads **145**, then the cross-cluster load balancer **139** can move to step **439**.

(68) In step **439**, the cross-cluster load balancer **139** can determine whether there are any additional source clusters **106** that have not been analyzed for cross-cluster migration. If there are additional source clusters **106**, then the cross-cluster load balancer **139** can move to step **403** or **406**. As a result, the cross-cluster load balancer **139** can iteratively analyze the set of participant clusters **106** to determine whether their workloads **145** can benefit from cross-cluster migration. If there are no source clusters **106**, then the cross-cluster load balancer **139** can move to step **427**.

(69) Stored in the memory device are both data and several components that are executable by the processor. Also stored in the memory can be a datastore **130** and other data. A number of software components are stored in the memory and executable by a processor. In this respect, the term “executable” means a program file that is in a form that can ultimately be run by the processor. Examples of executable programs can be, for example, a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of one or more of the memory devices and run by the processor, code that can be expressed in a format such as object code that is capable of being loaded into a random access portion of the one or more memory devices and executed by the processor, or code that can be interpreted by another executable program to generate instructions in a random access portion of the memory devices to be executed by the processor. An executable program can be stored in any portion or component of the memory devices including, for example, random access memory (RAM), read-only memory (ROM), hard drive, solid-state drive, USB flash drive, memory card, optical disc such as compact disc (CD) or digital versatile disc (DVD), floppy disk, magnetic tape, or other memory components.

(70) Memory can include both volatile and nonvolatile memory and data storage components. In addition, a processor can represent multiple processors and/or multiple processor cores, and the one or more memory devices can represent multiple memories that operate in parallel processing circuits, respectively. Memory devices can also represent a combination of various types of storage devices, such as RAM, mass storage devices, flash memory, or hard disk storage. In such a case, a local interface can be an appropriate network that facilitates communication between any two of the multiple processors or between any processor and any of the memory devices. The local interface can include additional systems designed to coordinate this communication, including, for example, performing load balancing. The processor can be of electrical or of some other available construction.

(71) Client devices **108** can be used to access user interfaces generated to configure or otherwise interact with the cluster management service **135**. These client devices **108** can include a display upon which a user interface generated by a client application for providing a virtual desktop session (or other session) can be rendered. In some examples, the user interface can be generated using user interface data provided by the management environment **103**. The client device **108** can also include one or more input/output devices that can include, for example, a capacitive touchscreen or other type of touch input device, fingerprint reader, or keyboard.

(72) Although the cluster management service **135** and other various systems described herein can be embodied in software or code executed by general-purpose hardware as discussed above, as an alternative the same can also be embodied in dedicated hardware or a combination of software/general purpose hardware and dedicated hardware. If embodied in dedicated hardware, each can be implemented as a circuit or state machine that employs any one of or a combination of a number of technologies. These technologies can include discrete logic circuits having logic gates for implementing various logic functions upon an application of one or more data signals, application specific integrated circuits (ASICs) having appropriate logic gates, field-programmable gate arrays (FPGAs), or other components.

(73) The sequence diagram and flowcharts show an example of the functionality and operation of an implementation of portions of components described herein. If embodied in software, each

block can represent a module, segment, or portion of code that can include program instructions to implement the specified logical function(s). The program instructions can be embodied in the form of source code that can include human-readable statements written in a programming language or machine code that can include numerical instructions recognizable by a suitable execution system such as a processor in a computer system or other system. The machine code can be converted from the source code. If embodied in hardware, each block can represent a circuit or a number of interconnected circuits to implement the specified logical function(s).

(74) Although the sequence diagram flowcharts show a specific order of execution, it is understood that the order of execution can differ from that which is depicted. For example, the order of execution of two or more blocks can be scrambled relative to the order shown. In addition, two or more blocks shown in succession can be executed concurrently or with partial concurrence. Further, in some examples, one or more of the blocks shown in the drawings can be skipped or omitted.

(75) Also, any logic or application described herein that includes software or code can be embodied in any non-transitory computer-readable medium for use by or in connection with an instruction execution system such as, for example, a processor in a computer system or other system. In this sense, the logic can include, for example, statements including program code, instructions, and declarations that can be fetched from the computer-readable medium and executed by the instruction execution system. In the context of the present disclosure, a “computer-readable medium” can be any medium that can contain, store, or maintain the logic or application described herein for use by or in connection with the instruction execution system.

(76) The computer-readable medium can include any one of many physical media, such as magnetic, optical, or semiconductor media. More specific examples of a suitable computer-readable medium include solid-state drives or flash memory. Further, any logic or application described herein can be implemented and structured in a variety of ways. For example, one or more applications can be implemented as modules or components of a single application. Further, one or more applications described herein can be executed in shared or separate computing devices or a combination thereof. For example, a plurality of the applications described herein can execute in the same computing device, or in multiple computing devices.

(77) It is emphasized that the above-described examples of the present disclosure are merely possible examples of implementations set forth for a clear understanding of the principles of the disclosure. Many variations and modifications can be made to the above-described embodiments without departing substantially from the spirit and principles of the disclosure. While aspects of the disclosure can be described with respect to a specific figure, it is understood that the aspects are applicable and combinable with aspects described with respect to other figures. All such modifications and variations are intended to be included herein within the scope of this disclosure.

Claims

1. A non-transitory computer-readable medium comprising program instructions that when executed cause at least one computing device to at least: identify a set of clusters to analyze for cross-cluster workload migration; generate a ranked list of the set of clusters; identify a source cluster for cross-cluster migration based on the ranked list; determine a cross-cluster migration gain to migrate a workload from the source cluster to a destination cluster, wherein the cross-cluster migration gain is determined based at least in part on a gain rate and a gain duration, the gain rate being based at least in part on a destination resource efficiency of the workload on the destination cluster, wherein the gain duration is an estimated duration of the gain rate at the destination cluster and is based at least in part on a cross-cluster migration time and a frequency of cross-cluster workload migrations to and from the destination cluster; generate a cross-cluster migration recommendation to migrate the workload from the source cluster to the destination cluster based at

least in part on the cross-cluster migration gain; determine that a maximum quantity of cross-cluster migration recommendations has been met based on a comparison between a total quantity of cross-cluster migration recommendations and the maximum quantity of cross-cluster migration recommendations, the cross-cluster migration recommendation being included to the total quantity of cross-cluster migration recommendations; and execute a migration of the workload from the source cluster to the destination cluster based at least in part on the maximum quantity of cross-cluster migration recommendations being met.

2. The non-transitory computer-readable medium of claim 1, wherein a respective cluster of the set of clusters comprises a storage area network that aggregates storage available to physical resources of the respective cluster.

3. The non-transitory computer-readable medium of claim 1, wherein the workload is identified based on a hardware resource requirement or an execution time.

4. The non-transitory computer-readable medium of claim 1, wherein the program instructions that when executed further cause the at least one computing device to at least: specify an optimal host in the destination cluster for the migration of the workload.

5. The non-transitory computer-readable medium of claim 1, wherein the destination cluster is selected based on the destination cluster having a lower cluster resource usage than the source cluster.

6. A system, comprising: at least one computing device; and program instructions stored in a datastore and executable in the at least one computing device that, when executed by the at least one computing device, cause the at least one computing device to at least: generate a ranked list of a set of clusters for cross-cluster workload migration; identify a source cluster for cross-cluster migration based on the ranked list; determine a cross-cluster migration gain to migrate a workload from the source cluster to a destination cluster, wherein the cross-cluster migration gain is determined based at least in part on a gain rate and a gain duration, the gain rate being based at least in part on a destination resource efficiency of the workload on the destination cluster, wherein the gain duration is an estimated duration of the gain rate at the destination cluster and is based at least in part on a cross-cluster migration time and a frequency of cross-cluster workload migrations to and from the destination cluster; generate a cross-cluster migration recommendation to migrate the workload from the source cluster to the destination cluster based at least in part on the cross-cluster migration gain; determine that a maximum quantity of cross-cluster migration recommendations has been met based on a comparison between a total quantity of cross-cluster migration recommendations and the maximum quantity of cross-cluster migration recommendations, the cross-cluster migration recommendation being included to the total quantity of cross-cluster migration recommendations; and execute a migration of the workload from the source cluster to the destination cluster based at least in part on the maximum quantity of cross-cluster migration recommendations being met.

7. The system of claim 6, wherein a respective cluster of the set of clusters comprises a storage area network that aggregates storage available to physical resources of the respective cluster.

8. The system of claim 6, wherein the workload is identified based on a hardware resource requirement or an execution time.

9. The system of claim 6, wherein the program instructions that when executed further cause the at least one computing device to at least: specify an optimal host in the destination cluster for the migration of the workload.

10. The system of claim 6, wherein the destination cluster is selected based on the destination cluster having a lower cluster resource usage than the source cluster.

11. The system of claim 6, wherein the program instructions, when executed by the at least one computing device, cause the at least one computing device to at least: determine that the migration of the workload fails based on at least one of a threshold migration time period being met, the destination cluster goes offline, or a cluster datastore goes offline.

12. The system of claim 11, wherein the program instructions, when executed by the at least one computing device, cause the at least one computing device to at least: delete a partially-deployed workload associated with the migration of the workload from the destination cluster based on the determination of the failure of the migration of the workload.
 13. The system of claim 6, wherein the ranked list of the set of clusters is generated based at least in part on a cluster resource usage of a respective cluster of the set of clusters.
 14. A computer-implemented method, comprising: generating a ranked list of a set of clusters for cross-cluster workload migration; identifying a source cluster for cross-cluster migration based on the ranked list; determining a cross-cluster migration gain to migrate a workload from the source cluster to a destination cluster, wherein the cross-cluster migration gain is determined based at least in part on a gain rate and a gain duration, the gain rate being based at least in part on a destination resource efficiency of the workload on the destination cluster, wherein the gain duration is an estimated duration of the gain rate at the destination cluster and is based at least in part on a cross-cluster migration time and a frequency of cross-cluster workload migrations to and from the destination cluster from the source cluster; generating a cross-cluster migration recommendation to migrate the workload from the source cluster to the destination cluster based at least in part on the cross-cluster migration gain; determining that a maximum quantity of cross-cluster migration recommendations has been met based on a comparison between a total quantity of cross-cluster migration recommendations and the maximum quantity of cross-cluster migration recommendations, the cross-cluster migration recommendation being included to the total quantity of cross-cluster migration recommendations; and executing a migration of the workload from the source cluster to the destination cluster based at least in part on the maximum quantity of cross-cluster migration recommendations being met.
 15. The computer-implemented method of claim 14, wherein a respective cluster of the set of clusters comprises a storage area network that aggregates storage available to physical resources of the respective cluster.
 16. The computer-implemented method of claim 14, wherein the workload is identified based on a hardware resource requirement or an execution time.
 17. The computer-implemented method of claim 14, wherein for the migration of the workload, specifying an optimal host in the destination cluster.
 18. The computer-implemented method of claim 14, wherein the destination cluster is selected based on the destination cluster having a lower cluster resource usage than the source cluster.
 19. The computer-implemented method of claim 15, further comprising: determining that the migration of the workload fails based on at least one of a threshold migration time period being met, the destination cluster goes offline, or a cluster datastore goes offline.
 20. The computer-implemented method of claim 19, further comprising: deleting a partially-deployed workload associated with the migration of the workload from the destination cluster based on the determination of the failure of the migration of the workload.
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